

Systematic Vital Signs Detection Framework Based on Frequency-Modulated Continuous Wave MIMO Radar

Yong Wang , Member, IEEE, Heng Liu , Wei Xiang , Senior Member, IEEE, Jiacheng Wang , Mu Zhou , Senior Member, IEEE, Yu Pang , and Dusit Niyato , Fellow, IEEE

Abstract—The frequency-modulated continuous wave (FMCW) radar has received much attention in the field of noncontact vital signs monitoring. However, since vital signs are usually very weak, it can be easily buried by interference and noise, especially for the heartbeat signal. To tackle this challenge, this article proposes a novel systematic vital signs detection framework using the multiple-input multiple-output FMCW radar. First, the signal noise ratio of the vital signs signal is enhanced by combining the phase signals of multiple channels using the maximum ratio combining method. Then, to suppress noise and interference, we construct the vital signs signal with singular spectral analysis and propose a correlation-based selection criterion to select potential intrinsic mode functions of the respiration and heartbeat signals. Finally, a fast independent component analysis is applied to extract the respiration signal, and the second-order derivative based fast independent component analysis in conjunction with an infinite impulse response notch filter is further developed to extract the heartbeat signal. Simulations and experimental results validate the effectiveness of the proposed framework.

Index Terms—Frequency-modulated continuous wave, heartbeat, multiple-input–multiple-output, respiration.

I. INTRODUCTION

WITH the development of radar technology, frequency-modulated continuous wave (FMCW) radar has been applied in industrial fields for personnel positioning, behavior recognition, and vital signs detection. Respiration and heartbeat are fundamental vital signs, and they are important for medical research and health monitoring [1]. The physiological information can be used for medical diagnosis, health monitoring, and emotion recognition of the human body [2]. Currently, conventional respiration and heartbeat detection methods rely predominantly on wearable devices or contact electrodes [2]. However, for special groups, such as burn and scald patients, skin allergy patients, and infants, contact sensors are not suitable for the detection of vital signs. By contrast, the millimeter wave radar-based vital signs detection method can detect human respiration and heartbeat signals in a noncontact way, which has broad application prospects [3], [4], [5], [6]. Moreover, the wide bandwidth of the millimeter wave radar allows for very high-resolution data acquisition, enabling accurate respiration and heartbeat signal detection. Recently, the FMCW radar has been widely developed in various scenarios, which mainly includes vital signs detection in the medical field [3], gesture recognition [4], behavior recognition in human–computer interaction [5], and imaging radar in smart transportation [6]. This article focuses on respiration and heartbeat signal detection using the FMCW radar.

Conventional methods for respiration and heartbeat extraction based upon millimeter wave radar-based vital signs detection are mainly divided into two categories, namely, spectral analysis methods [7] and modal decomposition methods [8], [9], [10]. Specifically, the spectral analysis method separates the respiration signal, heartbeat signal, and clutter by bandpass filtering, and then combines them with the Fourier transform to obtain the respiration rate (RR) and heartbeat rate (HR) [7]. Although this method can obtain the RR and HR, the obtained results can be easily adversely affected by surrounding noise. On the other hand, the widely used modal decomposition methods are empirical mode decomposition (EMD) [8], ensemble empirical mode

Received 12 August 2024; accepted 15 February 2025. Date of publication 8 April 2025; date of current version 13 June 2025. This work was supported in part by the Science and Technology Research Program of Chongqing Education Commission under Grant KJQN202200636, in part by the Chongqing Natural Science Foundation Project under Grant CSTB2023NSCQ-MSX0832, Grant CSTB2023NSCQ-LZX0107, Grant CSTB2023NSCQ-MSX0249, and Grant CSTB2023NSCQ-LZX0014, and in part by the National Natural Science Foundation of China under Grant 52302059 and Grant 61901076. Paper no. TII-24-4092. (Corresponding author: Wei Xiang.)

Yong Wang is with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China, and also with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100049, China (e-mail: yongwang@cqupt.edu.cn).

Heng Liu and Mu Zhou are with the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: s210131127@stu.cqupt.edu.cn; zhoulmu@cqupt.edu.cn).

Wei Xiang is with the School of Computing, Engineering and Mathematical Sciences, La Trobe University, Melbourne, VIC 3086, Australia, and also with the College of Science and Engineering, James Cook University, Cairns, QLD 4870, Australia (e-mail: w.xiang@latrobe.edu.au).

Jiacheng Wang and Dusit Niyato are with the College of Computing and Data Science, Nanyang Technological University, Singapore 639798 (e-mail: jiacheng.wang@ntu.edu.sg; dniyato@ntu.edu.sg).

Yu Pang is with the School of Optoelectronics Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: pangyu@cqupt.edu.cn).

Digital Object Identifier 10.1109/TII.2025.3545053

1941-0050 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See <https://www.ieee.org/publications/rights/index.html> for more information.

decomposition (EEMD) [9], and variational mode decomposition (VMD) [10]. The respiration and heartbeat signals can be obtained by decomposing the modal signal within the frequency ranges. However, most modal decomposition methods do not consider the effect of respiration harmonics on the heartbeat signal.

In addition to the aforementioned methods, some recent studies propose improved methods for vital signs detection. In [11], a method based on compressed sensing is proposed to separate and reconstruct the heartbeat and respiration signals using compressive sensing based on the orthogonal matching pursuit (CS-OMP), which effectively improves the precision of detecting respiration and heartbeat signals. However, this method only works well at the distance of 0.76 m, and the robustness at different distances and angles are not investigated. Ling et al. [12] proposed to extract the heartbeat signal using an adaptive trap filter in combination with an empirical wavelet transform, which estimates HR by using the relationship between respiration harmonics and weights. However, when the noise and interference are severe, using a notch filter for interference suppression is ineffective. In [13], a VMD wavelet intervallic threshold segmentation method is presented to reduce noise in vital signs detection, and the method achieves relatively accurate separation results at different distances. However, the detection performance is easily affected by the low signal-to-noise ratio (SNR) that caused by the clutter interference and noise. López et al. [14] developed a multiple-input–multiple-output (MIMO) radar-based respiration detection method for vehicle occupants, which extracts displacement changes in the divided six human torso regions and obtains the RR in each region by combining peak searching with a bandpass filter. However, the interference and noise in the signal are not preprocessed, and the hardware cost is very high because they use six radar sensors. Arsalan et al. [15] proposed to combine a bandpass filter and a Kalman filter to estimate the HR, which suppress noise in the signal by continuously updating the passband range of the filter to obtain a more accurate heartbeat signal. However, this method requires a rough estimate of the initial HR, and if the rough estimate is inaccurate, all subsequent steps are meaningless, leading to poor robustness. Liu et al. [16] designed an mmRH system to extract the RR and HR by mapping the sparse spectra of the vital signs to an adaptive filter. However, due to individual differences, health status, environmental conditions, etc., the spectra of the vital sign signals may vary and the sparsity assumption may not be satisfied. As a result, the presented method may be unable to effectively extract an accurate RR and HR. Wang et al. [17] combined the singular spectrum analysis and improved complete EEMD to reconstruct the respiratory and heartbeat signals. They only test the performance of their method with human body keep static. Moreover, the influence of respiration harmonics on the heartbeat signal is not discussed.

Due to the clutter interference, noise, and respiration harmonics, the low SNR makes the above studies suffer from inaccurate vital signs detection and poor robustness. Therefore, we propose a systematic detection framework using the FMCW

MIMO radar to extract respiration and heartbeat signals. The main contributions of this article are as follows.

- 1) We propose to apply a maximum ratio combining (MRC) method to enhance SNR of both human body and vital signs signals by combining the phase signals of multiple virtual antennas and the phase signals of multiple virtual antennas using the weight vector obtained from eigenvalue decomposition.
- 2) We suppress noise and interference by selecting the singular components of singular spectral analysis with key cumulative energy and propose a correlation-based selection criterion to select potential intrinsic mode functions (IMFs) for respiration and heartbeat signals construction.
- 3) We apply the fast independent component analysis (FastICA) to extract the respiration signal. The second-order derivative in conjunction with an infinite impulse response (IIR) notch filter FastICA (SOD-FICA) is further proposed to eliminate the respiratory harmonics and extract the heartbeat signal.
- 4) We carry out both simulations and experiments to verify the effectiveness of our framework. The results indicate that the RR and HR errors of our framework are smaller than the comparison methods. Moreover, the experiments conducted at different distances, angles and body movement also prove the robustness of proposed framework.

The rest of this article is organized as follows. The respiration and heartbeat model are described Section II. Section III describes the detailed proposed framework. Section IV validates our method by simulation and real data. Finally, Section V concludes this article.

II. MULTIANTENNA FMCW RADAR SIGNAL MODEL

The FMCW radar sensor consists of several modules, such as the radio frequency (RF), analog clock, and digital circuitry [18]. The transmit antenna of the FMCW radar sensor emits a linear frequency-modulated (FM) signal the human chest cavity reflects the FM signal. Then, the reflected signal and the transmitted signal are mixed. The intermediate frequency (IF) signal is obtained from the mixed signal using a low-pass filter. Finally, the IF signal is sampled by an analog-to-digital converter (ADC), and the digital signal is sent to the digital signal processing module for further processing.

Since the FMCW radar usually contains multiple antennas, a time-division multiplexing approach is adopted [18]. Denoting by M and N , respectively, the numbers of transmit and receive antennas, the transmit signal of the m th antenna is expressed as follows:

$$s_m(t) = A_m \exp(j(2\pi(f_c(t - (m-1)T_d) + \frac{K}{2}(t - (m-1)T_d)^2) + \varphi_m(t))) \quad (1)$$

where A_m is the amplitude of the transmit signal of the m th transmit antenna, f_c is the starting frequency of the transmit signal, K denotes the slope of the linear FM signal, T_d is the

time interval between the transmit antennas, and $\varphi_m(t)$ is the initial phase of the m th transmit antenna.

Accordingly, the movements of the respiration and heartbeat can be modeled as a sinusoidal signal [19]

$$y(t) = A_r \sin(2\pi f_r t) + A_h \sin(2\pi f_h t) \quad (2)$$

where A_r and A_h are, respectively, the amplitudes of the respiration and heartbeat movements, and f_r and f_h are the corresponding frequencies.

Denote by d_0 the distance between the chest and the radar, and c the light speed. The latency between the transmit and receive signals is $t_d = 2(d_0 + y(t))/c$. With multiple transmitters and receivers, virtual antenna channels of the phase signals can be generated. The distance between adjacent receive antennas is d , while the distance between the two transmit antennas is $4d$. Assuming that θ is the angle between the human chest and receive antenna, then the wave path difference between the virtual antenna formed by the m th transmit and the n th receive antennas is $d_{mn} \sin \theta$. Therefore, the signal of each receive antenna can be expressed as

$$r_{mn}(t) = A_{mn} \exp \left(j \left(2\pi \left(f_c(t - (m-1)T_d - t_d) + \frac{K}{2}(t - (m-1)T_d - t_d)^2 + \frac{d_{mn} \sin \theta}{\lambda} \right) + \varphi_{mn}(t) \right) \right) \quad (3)$$

where A_{mn} is the amplitude of the received signal from the m th transmit antenna to the n th virtual receive antenna, and λ is the wavelength of the signal [20]. The transmit and receive signals are orthogonally mixed and filtered out the high frequency part. Then, the IF signal is expressed as

$$s_{IF}(t) = A_m A_{mn} \exp \left(j \left(2\pi \left(f_c t_d + K t_d t - K(m-1)T_d t_d - \frac{K}{2} t_d^2 - \frac{d_{mn} \sin \theta}{\lambda} \right) + \varphi_m(t) - \varphi_{mn}(t) \right) \right). \quad (4)$$

We substitute $t_d = 2(d_0 + y(t))/c$ and $f_c = \frac{c}{\lambda}$ into (4). $K(m-1)T_d t_d$, $\frac{K}{2} t_d^2$, $\varphi_m(t) - \varphi_{mn}(t)$, and $2K y(t)t/c$ are ignored because of their extremely small values [11]. Therefore, (4) can be approximated as

$$s_{IF}(t) \approx A_m A_{mn} \exp \left(j \left(\frac{4\pi(d_0 + y(t))}{\lambda} + \frac{4\pi K d_0}{c} t - \frac{2\pi d_{mn} \sin \theta}{\lambda} \right) \right). \quad (5)$$

We learn from (5) that for a given distance between a human body and the radar, the phase variation in (5) is expressed by the vibration displacement $\Delta\psi = \frac{4\pi}{\lambda} y(t)$. The phase signal primarily contains the displacement variations caused by the respiration and heartbeat movements. Moreover, it includes respiration harmonics, noise, as well as the micromotion components. Because of the small displacement changes of the heartbeat, the heartbeat motion reflected in the phase is also very weak. Moreover, in the presence of noise and body shaking, it is difficult to extract respiration and heartbeat signals.

III. PROPOSED METHOD

A. Framework Overview

To implement radar-based vital signs detection, we present a three-step framework for vital signs detection.

Step 1—Human chest localization and phase extraction: The fast fourier transform (FFT) is carried out on each IF signal from the virtual antennas to acquire the range-time map, and then, the phase signals at the range of the human chest are extracted. Next, the phase signals of the virtual antennas are merged by using the proposed MRC to enhance the SNR of the phase signals from the human chest.

Step 2—Interference and noise reduction: We employ the singular spectrum analysis algorithm to suppress micro-motion interference and noise in the phase signal. Then, the phase signal is decomposed into multiple IMFs by using complete ensemble empirical mode decomposition with adaptive noise (CEEMDAN) [21]. We design a correlation-based IMF selection criterion to effectively select the potential respiration and heartbeat signals.

Step 3—Vital signs signal extraction: The FastICA is adopted to obtain the respiration signals. Then, the SOD-FICA is further proposed to eliminate the effect of respiration harmonics on heartbeat signal and accurately extract heartbeat signal.

B. Human Chest Localization and Phase Signal Extraction

To realize vital signs detection, it is necessary to observe the displacement changes over multiple sweep periods. After the IF signal is sampled by ADC, it will form a 2-D matrix, where the column indicates the chirps, while the row indicates the sampling points of each chirp. According to (5), the frequency of the IF signal is determined by the distance between the human chest and radar. By performing the FFT on each chirp, the position of the human chest can be obtained, and the corresponding orthogonal baseband signals can be extracted. Because the displacement caused by respiration is multiple times the wavelength, the phase of the demodulation is beyond the range of $(-\pi/2, \pi/2)$. To recover the phase due to respiration, we use inverse tangent demodulation to resolve such a phase ambiguity problem [11]. Although we can use a single transmit and a single receive antenna to detect the vital signs, it is easily interfered by the interference and noise. Therefore, we use the phase signals of multiple virtual antennas to enhance the SNR and the detection performance [11]. Given the phase signals of multiple virtual antennas, we first construct the phase signal matrix $\Psi(\mathbf{t}) = [\psi_{11}(t), \psi_{12}(t), \dots, \psi_{24}(t)]^T$, where $\psi_{mn}(t)$ is the phase signal obtained from the m th transmit antenna to the n th receive antenna, and T is the transpose operator.

Different from [11] (sum multiple phase signals directly), we propose MRC to fuse multiple phase signals in an effort to improve the SNR. MRC merges the phase signals from multiple virtual channels to improve the quality and reliability of the phase signal [22]. Specifically, to combine the signals, we first construct the following correlation matrix:

$$C_{ss} = \Psi(\mathbf{t})\Psi^H(\mathbf{t}) \quad (6)$$

where H denotes the transpose conjugate operator. Denote by P the number of eigenvectors obtained after matrix decomposition. The matrix eigenvalue decomposition yields

$$\mathbf{C}_{ss} = [\mathbf{v}_1, \dots, \mathbf{v}_P] \text{diag}\{\sigma_1, \dots, \sigma_P\} [\mathbf{v}_1, \dots, \mathbf{v}_P]^H \quad (7)$$

where $\sigma_1, \dots, \sigma_P$ are the eigenvalues in descending order, and $\mathbf{v}_1, \dots, \mathbf{v}_P$ are the eigenvectors corresponding to each eigenvalue, for each of which there is $|\mathbf{v}_q|^2 = 1 (q = 1, 2, \dots, P)$.

Then, we use the eigenvector corresponding to the maximum eigenvalue as a weight vector to combine the phase signals of multiple virtual antennas. The fused phase signal is obtained as $x(t) = \mathbf{v}_1^H \Psi(\mathbf{t})$. Then, the fused phase signal will be further processed for vital signs detection in the next subsection. In the experiments, we have verified the effect of MRC and found that the MRC operation not only helps us to find the correct position of human body by suppressing the clutter interference, but also enhance the SNR of both respiration and heartbeat signals. Furthermore, the proposed framework can be extended to multihuman vital signs detection by using the beamforming technology to find different human chests. However, more effort should be made and we leave it for future work.

C. Interference and Noise Suppression

In this section, we will denoise the fused signal and reduce the interference.

1) *SSA-based interference and noise reduction*: The merged signal contains the human respiration and heartbeat signals, as well as various interference and noise. To suppress the interference and noise, we employ the SSA to decompose the phase signal and reconstruct the signal by partial eigenvalues. The SSA algorithm mainly consists of four steps, namely, trajectory matrix construction, singular value decomposition (SVD), eigenvalue selection, and phase signal reconstruction [23]. In the process of SVD, the singular value matrix is obtained in descending order, and the magnitude of each singular value indicates the energy or importance of each component. Therefore, it is crucial to select important singular values for signal reconstruction. In this article, an energy preservation criterion is employed to retain significant constituents of the human thoracic phase signal. Specifically, the singular components are selected in descending order for signal reconstruction until the cumulative energy reaches 95% of the total energy. The reconstructed signal $\mathbf{X}(t)$ preserves the primary vital signs contained in the phase signal while filtering out a significant portion of the noise and interference signals.

2) *Correlation-based IMF selection*: Although some interference and noise are suppressed, the heartbeat signal is easily overwhelmed by the respiration harmonics. Therefore, we propose a correlation-based IMF selection method to choose potential vital signs signals. We decompose the phase signal into multiple IMFs by CEEMDAN and design an IMF selection criterion to effectively choose potential respiration and heartbeat signals, while further reducing the noise and interference.

First, we apply CEEMDAN to decompose the vital signal $\mathbf{X}(t)$ into several IMFs [21]

$$\mathbf{X}(t) = \sum_{l=1}^L \text{IMF}_l(t) + r(t) \quad (8)$$

where L is the total number of IMFs, $\text{IMF}_l(t)$ is the l th IMF, and $r(t)$ is the residual component of the phase signal, which is a monotonic function. Although CEEMDAN alleviates the mode mixing problem, the decomposed IMFs still suffer from this problem. As a result, the respiration and heartbeat signals may be distributed across different IMFs. Therefore, these signals should be further extracted by selecting useful IMFs.

This article proposes an IMF selection criteria to choose the potential sets that contain the respiration and heartbeat signals, by calculating the proportion of components in the range of 0.1–0.5 Hz and 0.8–2.0 Hz within each IMF signal. First, we define the degree of correlation in each IMF signal for the respiration and heartbeat signals as

$$\alpha_j = \frac{E_r(j)}{E(j)}, \text{ and } \beta_j = \frac{E_h(j)}{E(j)} \quad (9)$$

where α_j and β_j are the degrees of correlation of the j th IMF signal for the respiration and heartbeat signals, respectively, $E(j)$ is the energy of the j th IMF signal, $E_r(j)$ is the energy within the frequency bands of respiration of the j th IMF, and $E_h(j)$ is the energy within the frequency bands of heartbeat of the j th IMF.

After calculating the degree of correlation of respiration and heartbeat in each IMF, we select the IMF with the highest correlation and its adjacent two IMFs to reconstruct the potential respiration signal $\mathbf{X}_r(t)$ and heartbeat signal $\mathbf{X}_h(t)$. Note that if the first IMF has the highest degrees of correlation, we choose the first two IMFs to reconstruct the respiration (or heartbeat) signal. The rest IMFs that contain interference and noise are excluded. The experiments in the following section also demonstrate that such a scheme can effectively improve the vital signs detection performance.

D. Vital Signs Signal Extraction

After interference and noise suppression, we try to extract respiration and heartbeat signals from the obtained IMFs in this section.

1) *FastICA-Based Respiration Signal Extraction*: ICA is a signals separation method that can extract the source signal from mixed data. According to Comon [24], if the mixed signal is a linear combination of independent source signals and exhibit non-Gaussian properties among the source signals, it can be used to perform blind source signal separation on the mixed signal. FastICA is known for its high computational efficiency and is suited to extract vital signs signal, as it satisfies the properties of independence and non-Gaussianity [25], [26]. Therefore, we employ the FastICA algorithm to extract the respiration signal. The potential respiration signal set $\mathbf{X}_r(t)$ can be expressed as

$$\mathbf{X}_r(t) = \mathbf{A}\mathbf{S}(t) \quad (10)$$

where $\mathbf{X}_r(t)$ and $\mathbf{S}(t)$ are the mixed signal matrix and the source signal matrix, respectively, and $\mathbf{A}$ is the mixing matrix. To find the source signal matrix $\mathbf{S}(t)$, we need to find the separation matrix $\mathbf{A}^{-1}$.

FastICA finds the weight vector by maximizing the non-Gaussianity of the respiration signal in the direction of the weight vector. The FastICA algorithm consists of three steps: 1) centering the data, 2) whitening the data, and 3) calculating the weight vector. More details of the FastICA algorithm can be found in [26]. After calculating all the weight vectors, we obtain the separation matrix $\mathbf{W}$, and $\mathbf{W}$ can be used to compute the independent source signals matrix $\mathbf{S} = \mathbf{W}\mathbf{z}$, where $\mathbf{z}$ is the matrix after whitening. Finally, we perform the FFT on the source signals, and the signal with the highest frequency peak is designated as the respiration signal.

2) SOD-FICA Algorithm for Heartbeat Signal Extraction: Compared with respiration, the heartbeat signal is much weaker. Moreover, the respiratory harmonics and interference can easily drown out the heartbeat signal. As a result, it is challenging to extract accurate heartbeat signal. In this section, we propose a novel SOD-FICA algorithm for heartbeat signal extraction. First, we model the respiration and heartbeat signals and their harmonics as periodic signals:

$$L(t) = \sum_{p=1}^{+\infty} A_{rp} \sin(2\pi p f_r t) + \sum_{q=1}^{+\infty} A_{hq} \sin(2\pi q f_h t) \quad (11)$$

where A_{rp} is the amplitude of the p th respiration harmonic signal, f_r is the corresponding frequency, A_{hq} is the amplitude of the q th heartbeat harmonic, and f_h is the corresponding heartbeat frequency. Then, the n th order derivative of $L(t)$ is

$$x^{(n)}(t) = \sum_{p=1}^{+\infty} A_{rp} (2\pi p f_r)^n \sin\left(2\pi p f_r t + \frac{n\pi}{2}\right) + \sum_{q=1}^{+\infty} A_{hq} (2\pi q f_h)^n \sin\left(2\pi q f_h t + \frac{n\pi}{2}\right). \quad (12)$$

According to (12), the ratio of the amplitudes of the heartbeat and respiration harmonics becomes

$$\varepsilon_p = \frac{(2\pi p f_h)^n A_h}{(2\pi p f_r)^n A_{rp}} = \frac{(f_h)^n A_h}{(p f_r)^n A_{rp}} = \left(\frac{f_h}{p f_r}\right)^n \frac{A_h}{A_{rp}}. \quad (13)$$

As can be seen from (13), the ratio of the heartbeat signal to the respiration harmonics increases with $(\frac{f_h}{p f_r})^n$. Therefore, by computing high-order derivatives of the potential heartbeat signals, we can increase the strength of the heartbeat relative to the respiratory harmonics. Typically, since the ratio between RR and HR is 1:4 [27], we can effectively reduce the impact of the second and third harmonics on the heartbeat signal by calculating the derivatives of the potential heartbeat signals. To suppress the respiration harmonics and reduce the computational effort, we utilize the second-order derivative of the signal.

On the other hand, to further eliminate higher-order harmonics of the respiration signal, we introduce an IIR notch filter to attenuate the respiration harmonics. Assuming that the amplitude

characteristic curve is approximately zero at the specified frequency point f_r (where f_r is the obtained respiration frequency). The trap digital frequency w_0 and the system function $H(z)$ are

$$w_0 = \frac{2\pi f_r}{f_s}, \text{ and } H(z) = \frac{1 - (2 \cos w_0)z^{-1} + z^{-2}}{1 - (2r \cos w_0)z^{-1} + r^2 z^{-2}} \quad (14)$$

where f_s is the sampling rate, r is the trap constant, and z is the complex variable. After obtaining the respiration frequency f_r by FastICA, we apply stopbands at integer multiples of f_r . The impact of respiration harmonics on the heartbeat is eliminated. Then, we employ the FastICA algorithm to extract the filtered heartbeat signal from the obtained multiple IMFs. The FFT is further performed on each produced independent source signal, and the highest spectral peak is selected as the heartbeat signal. The algorithm uses an iterative optimization method to find independent components, and thus is of low computational complexity.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. Simulation Analysis

To validate the feasibility of our proposed framework, we first carry out computer simulations. We set the respiration signal and two respiration harmonic signals. The amplitude of the main respiration signal is 1 at frequency 0.4 Hz. The amplitude of the second respiration harmonic is 0.2 at frequency 0.8 Hz, and the amplitude of the third respiration harmonic is 0.1 at frequency 1.2 Hz. The heartbeat signal has an amplitude of 0.1 at frequency 1.3 Hz, alongside the Gaussian white noise with zero mean and unit variance.

The simulated signal and its spectrum are shown in Fig. 1(a) and 1(b). After SSA processing and IMF selection, we perform FastICA on the potential respiration signal set, and the results are shown in Fig. 1(c) and 1(d), where the red line represents the reference frequency. As can be observed from Fig. 1, the second independent component corresponds to the highest spectral peak at 0.4 Hz, which is regarded as the respiration signal. We use the SOD-FICA algorithm to process the potential heartbeat signal set, and the results are shown in Fig. 1(e) and 1(f). Similarly, the red line in the figure is the accurate frequency. It can be observed that the second extracted signal corresponds to the highest spectral peak at 1.3 Hz, suggesting successful extraction of the heartbeat signal. Therefore, these simulation results validate the effectiveness of proposed framework.

B. Experimental Results and Analysis

In this section, we carry out experiments to verify the effectiveness of our method.

1) Experimental System Setup and Parameters Configuration: The radar equipment, same as in [17], is the AWR1642 radar sensor [28] equipped with the DCA1000 data acquisition card from Texas Instruments. The AWR1642 is used to transmit and receive signal, and the starting frequency is 77 GHz, and DCA1000 is used to capture raw data (IF signal) from the radar sensor. The AWR1642 contains two transmit and four receive antennas. The radar parameters can be adjusted by a PC, and

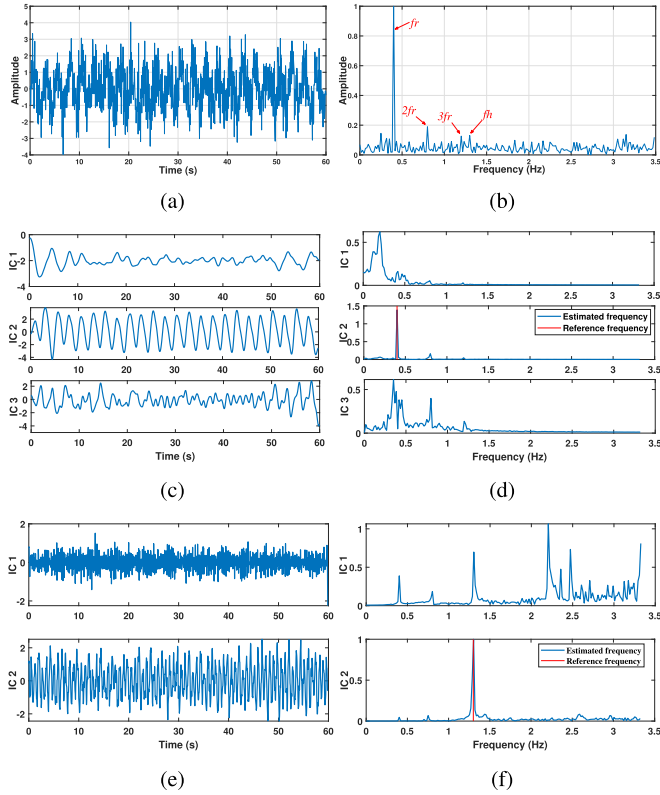


Fig. 1. Simulation results. (a) Simulated mixed signal. (b) Spectrum of the mixed signal. (c) Respiration signals after FastICA. (d) Spectrum of respiration after FastICA. (e) Heartbeat signals after SOD-FICA. (f) Spectrum of heartbeat after SOD-FICA.

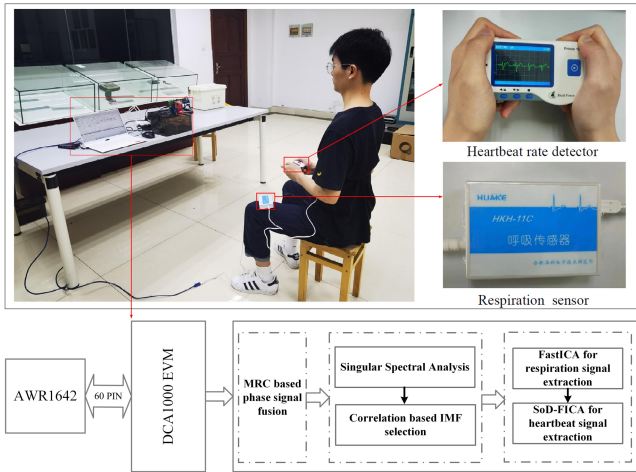


Fig. 2. Experimental scenarios and equipment.

the transmit and receive signals are multiplied and filtered to acquire IF signals, which are then transmitted to DCA1000 via cable. Finally, we transmit the original IF signal to the PC for signal processing through an Ethernet cable. The experimental scenario is shown in Fig. 2, where a human subject is seated in front of the radar sensor.

For the parameter settings of the radar sensor, larger bandwidth is typically required to achieve satisfactory detection

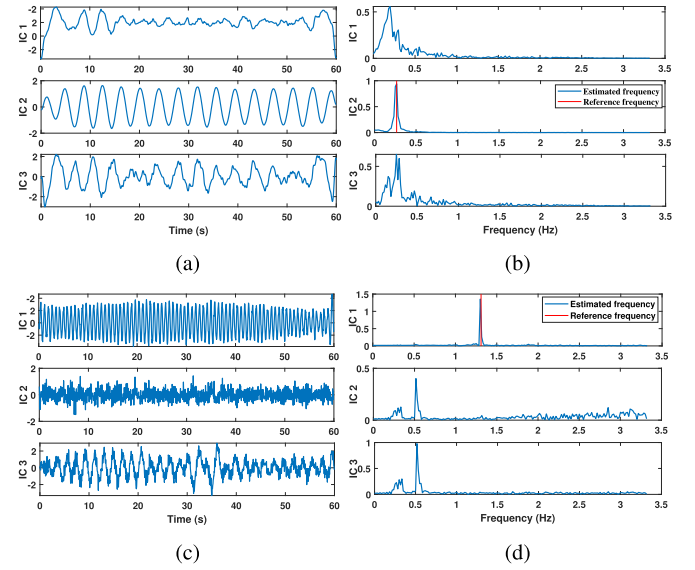


Fig. 3. Experimental results. (a) Respiration signals after FastICA. (b) Spectrum of respiration after FastICA. (c) Heartbeat signals after SOD-FICA. (d) Spectrum of heartbeat after SOD-FICA.

performance. Therefore, we set the slope of the frequency modulation signal to $70 \text{ MHz}/\mu\text{s}$, and the corresponding bandwidth is 3.99 GHz. The vital signs data are collected in 1 min, and we set the number of frames to 1200 and the frame length to 50 ms. We send only one sweep in each frame. The idle time and ramp end time of each chirp are 7 and $57 \mu\text{s}$, respectively. Each chirp consists of 200 sampling points, and the sampling rate is set to 4000 kps.

For performance comparison, we use the Prince 180B ECG detector from Likang Company to collect the reference HR [29] and the HKH-11 C respiration sensor from the Hefei Huake Electronic Technology Research Institute to collect the reference RR [30].

2) Detailed Results of the Proposed Framework: In the experiment, the human subject faces the radar and sits one meter away from it. The experimental results are shown in Fig. 3. We learn from Fig. 3(a) and 3(b) that the second independent component has the highest spectral peak at 0.25 Hz, which is almost equal to the reference respiration frequency. Similarly, since the heartbeat signal is enhanced by SOD-FICA, the first independent component has the highest spectral peak at the frequency of 1.32 Hz [see Fig. 3(c) and 3(d)], which is almost equal to the reference heartbeat frequency.

We also evaluate the SNR improvement of MRC operation using multiple antennas. In the experiment, we compare the SNRs of respiration and heartbeat signals using the fused phase signal and the final vital signs signal with data from the first virtual antenna. The SNR is calculated as $\text{SNR} = 10 \lg(E_{\text{signal}}/E_{\text{noise}})$, where E_{signal} is the energy of the respiration or heartbeat signal and E_{noise} is the energy of the interference plus noise. We randomly calculate the average SNR of five experiments as shown in Table I. Accordingly, MRC operation improves the SNRs of both respiration and heartbeat signals by about 2 and

TABLE I
SNR IMPROVEMENT OF THE MRC METHOD IN DB

Experiments	SNR of respiration signal		SNR of heartbeat signal	
	Single	MRC	Single	MRC
SNR of the final vital signs signal	7.33	11.46	6.01	10.27
SNR of the phase signals	-8.74	-6.42	-10.26	-8.18

4 dB for the fused phase signal and the final vital signs signal, respectively.

3) Performance Comparison: To validate the performance improvement of our framework, we collect data from forty individuals at the distance of one meter from the radar, including 25 males and 15 females.¹ The ages of the testers are 17 to 49 years old (averaged 26.7 years old), and their weights are 43–93 kg (averaged 60 kg), and their heights are 153–195 cm (averaged 168.8 cm). We use a HR monitor and a respiration sensor to obtain the reference values. Moreover, we compare our framework with four comparative methods, including the filtering method [7], EEMD algorithm [9], CS-OMP algorithm [11], mmRH [16], and VISION [17]. The statistical results are given in Table II, and the results are given in terms of beats per minute (BPM) for each method, mean absolute error (MAE), root mean squared error (RMSE), and mean relative error (MRE). As illustrated in Table II, the proposed framework has the lowest MAE (0.86 BPM) and RMSE (1.01 BPM) of respiration than the comparative methods, and the corresponding MRE is 4.46%. Similarly, the proposed framework achieves the lowest MAE (2.52 BPM) and RMSE (2.70 BPM) for heartbeat, and the corresponding MRE is 3.16%. This is because the proposed framework is capable of effectively suppress noise, interference, and respiration harmonics, so as to improve the detection accuracy of the heartbeat signal. In summary, our framework exhibits better performance compared to other methods.

Moreover, we compare the computational complexity and running time of the proposed framework with the existing alternatives. The complexity of the filtering method is $O(N)$, while computational complexities are EEMD, CS-OMP, and mmRH are $O(N^2)$, where N is the length of the time series of the vital signs signals. The computational complexity of our proposed framework mainly stems from SVD ($O(N^3)$), CEEMDAN ($O(IJN^2)$), and independent component analysis ($O(KN)$), where I is the iteration number of CEEMDAN, J is the times of EMD decomposition, and K is the number of mixed signals. Since I , J , and K are small (less than 10) [21], the computational complexity of the proposed framework can be briefly expressed as $O(N^3)$. The running time of 20 datasets with 5 s data are averaged and compared with VISION [17], and the results are given in Table III. It can be seen that the averaged running time of the proposed framework is slightly larger than

¹ Although we try to increase the test samples with different body type, much more samples are required to discuss the details impact of body types on the detection performance. In the next step, we will cooperate with hospital and test the detection performance of the proposed system with much more samples.

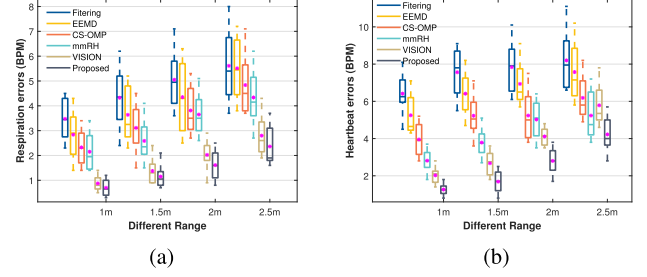


Fig. 4. Errors comparison at different ranges. (a) Respiration. (b) Heartbeat.

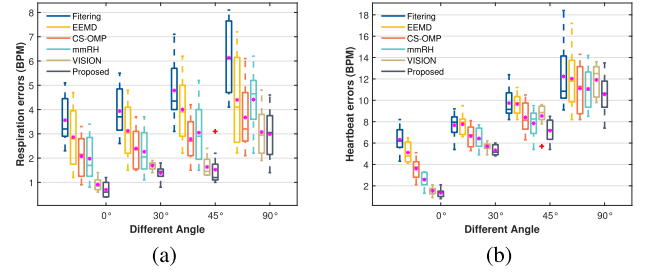


Fig. 5. Errors comparison at different horizontal angles. (a) Respiration. (b) Heartbeat.

that of the existing methods. Nevertheless, we can obtain better respiration and heartbeat performance (please see Table II), and the running time of our method supports real-time applications. It is noted that in real-time application, we compute the RR and HR with 5 s data and the rates are updated every second.

4) Effect of Distance: We investigate the performance at different distances. The error results of the RR and HR are compared in Fig. 4. It demonstrates that the errors of both the RR and HR increase with the distance away from the radar. Nevertheless, compared to other methods, the proposed one has a smaller error. On the other hand, since the heartbeat signal weakens with the increasing distance, the maximum heartbeat error increases from 1 to 5 BPM. Therefore, we can conclude that proposed framework can measure the RR and HR at short distances with high accuracy.

5) Effect of Horizontal Angle: We also compare proposed framework with alternative methods at different horizontal angles. In Fig. 5, the error rates of both respiration and heartbeat increase with the angle. It is interesting to observe that although the RR error becomes larger for larger angles, we can still measure the respiration when the angle is 90°. However, a significant error rate in the detected heartbeat is observed with larger angles, and we no longer can measure the heartbeat signal when the angle is 90°. This is because the phase variations from respiration are distinguishable, while the phase variations from heartbeat are too weak to be detected.

6) Effect of Clothing Material: We also analyze the effect of different clothing materials on the detection performance, shown in Fig. 6. In Fig. 6, we give the error results when wearing a T-shirt, a sweater and a down jacket, respectively. As illustrated in the figure that the clothing materials have effect

TABLE II
ERROR PARAMETERS COMPARISON OF RESPIRATION OBTAINED BY COMPARATIVE METHODS

Types of Vital Signs	Error Parameters	Filtering [7]	EEMD [9]	CS-OMP [11]	mmRH [16]	VISION [17]	Proposed
Respiration	MAE (BPM)	3.61	2.96	2.22	1.73	1.24	0.86
	RMSE (BPM)	3.71	3.06	2.29	1.83	1.33	1.01
	MRE (%)	18.83	15.59	11.67	9.02	6.34	4.46
Heartbeat	MAE (BPM)	8.83	6.43	4.58	4.02	3.16	2.52
	RMSE (BPM)	9.17	6.73	4.80	4.29	3.59	2.70
	MRE (%)	11.26	8.18	5.82	5.05	4.27	3.16

TABLE III
AVERAGE RUNNING TIME COMPARISON (MS)

Filtering [7]	EEMD [9]	CS-OMP [11]	mmRH [16]	VISION [17]	Proposed
18	87	99	116	146	151

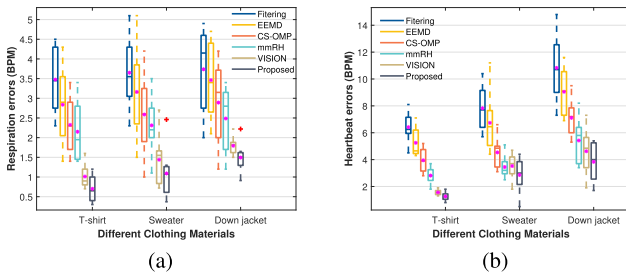


Fig. 6. Errors comparison at different clothing material. (a) Respiration. (b) Heartbeat.

on the extraction of breathing and heartbeat signals. The average error of respiration for our method is 1.5 BPM when wearing a down jacket, and 1.1 BPM when wearing a sweater, and 0.7 BPM when wearing a T-shirt. For the heartbeat, the average error of our method is 3.8 BPM for down jacket, 2.9 BPM for sweater, and 1.3 BPM for T-shirt. This is because the T-shirt is thin, it absorbs less and reflects strong signals, and as a result the estimation errors of respiration and heartbeat are relatively small. The down jacket, on the other hand, is too thick and it may absorb more radar signals, as a result, the estimation errors of both respiration and heartbeat for the down jacket become larger. Nevertheless, the results in Fig. 6 demonstrate that the clothes materials slightly impact both the detection accuracy of respiration and heartbeat.

7) Effect of Elevation Angle: We also carry out experiments with different elevation angles and the error results are shown in Fig. 7. When the elevation angle increases from 0° to 30° , the average error of respiration gradually increases from 0.7 to 2.1 BPM, while the average error of heartbeat increases from 1.3 to 10.1 BPM. The reason is that although both human body's respiration and heartbeat cause the movement of the chest, when the elevation angle increases, the movement in the radial direction caused by respiration and heartbeat decreases. Since the respiratory tendency is much stronger than that of the heartbeat, the respiratory motion can be still detected (but performance deterioration) when the elevation angle increases from 0° to 30° , while radial heartbeat movements are sometimes too faint to be accurately detected.

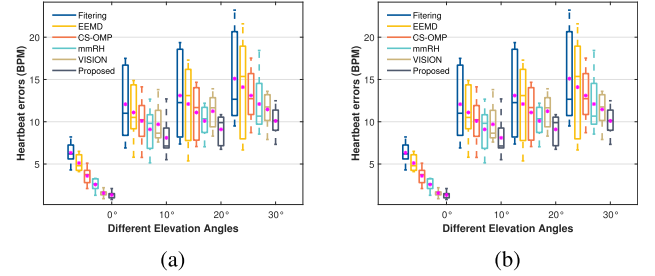


Fig. 7. Errors comparison at different elevation angles. (a) Respiration. (b) Heartbeat.

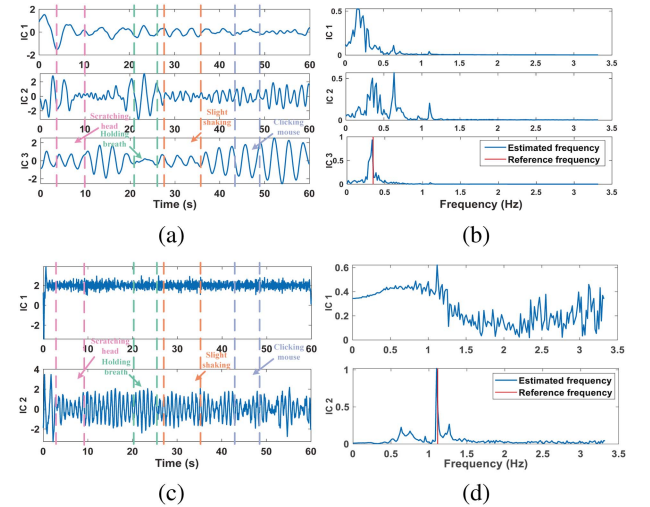


Fig. 8. Robustness test with slight human movement. (a) Respiration signals after FastICA. (b) Spectrum of respiration after FastICA. (c) Heartbeat signals after SOD-FICA. (d) Spectrum of heartbeat after SOD-FICA.

8) Robustness Test: To further validate the robustness of our framework, we conduct experiments with the human body moving around. In the experiment, the movements of the human body include “scratching the head,” “holding breath,” “slight shaking,” and “clicking a mouse.” The respiration and heartbeat signals as well as the corresponding spectra are shown in Fig. 8. The reference frequencies for respiration and heartbeat are 0.35 and 1.12 Hz, respectively. The obtained respiration and heartbeat signals are severely impaired in Fig. 8. However, the RR and HR are close to the reference ones, which show the robustness of the proposed framework with slight limb motions. This is because the proposed framework not only improves

the SNR, and suppresses the noise and interference, but also eliminates the respiration harmonics on the heartbeat signal.

V. CONCLUSION

In this article, a novel systematic vital signs signal detection framework using FMCW radar was proposed. First, we proposed to improve the SNR using MRC for multiple phase signals of virtual antennas. Then, we applied the SSA to reduce interference and noise and proposed a correlation-based IMF selection criterion to reconstruct potential respiration and heartbeat signals. Finally, we employed the FastICA to detect the respiration signal and proposed SOD-FICA to extract accurate heartbeat signals. Both simulation and experimental results were presented to indicate that the MAE enhancements of respiration and heartbeat for the proposed framework were, respectively, 0.74 and 2.34 BPM, and the RMSE enhancements of respiration and heartbeat for the proposed framework were, respectively, 0.79 and 2.37 BPM. Moreover, the robustness of the proposed framework was validated at various distances and angels, and with slight body movements. In the future, we will investigate the applicable of the proposed framework in more complex scenarios with multiperson and walking or running states, and strive to enhance the detection range as well as the accuracy.

REFERENCES

- [1] J. Liu, Y. Li, C. Li, C. Gu, and J. Mao, "Accurate measurement of human vital signs with linear FMCW radars under proximity stationary clutters," *Trans. Biomed. Circuits Syst.*, vol. 15, no. 6, pp. 1393–1404, Dec. 2021.
- [2] L. Zhang et al., "Non-contact dual-modality emotion recognition system by CW radar and RGB camera," *IEEE Sensors J.*, vol. 21, no. 20, pp. 23198–23212, Oct. 2021.
- [3] M. Mercuri et al., "2-D localization, angular separation and vital signs monitoring using a SISO FMCW radar for smart long-term health monitoring environments," *IEEE Internet Things J.*, vol. 8, no. 14, pp. 11065–11077, Jul. 2021.
- [4] Y. Wang, A. Ren, M. Zhou, W. Wang, and X. Yang, "A novel detection and recognition method for continuous hand gesture using FMCW radar," *IEEE Access*, vol. 8, pp. 167264–167275, 2020.
- [5] C. Ding et al., "Inattentive driving behavior detection based on portable FMCW radar," *IEEE Trans. Microw. Theory Techn.*, vol. 67, no. 10, pp. 4031–4041, Oct. 2019.
- [6] H. Song and H. Shin, "Classification and spectral mapping of stationary and moving objects in road environments using FMCW radar," *IEEE Access*, vol. 8, pp. 22955–22963, 2020.
- [7] U. Pangerc and F. Jager, "Robust detection of heart beats in multimodal records using slope-and peak-sensitive band-pass filters," *Physiol. Meas.*, vol. 36, no. 8, pp. 1645–1664, May 2015.
- [8] F. Weishaupt, I. Walterscheid, O. Biallawons, and J. Klare, "Vital sign localization and measurement using an LFMCW MIMO radar," in *Proc. 19th Int. Radar Symp.*, 2018, pp. 1–8.
- [9] L. Sun et al., "Remote measurement of human vital signs based on joint-range adaptive EEMD," *IEEE Access*, vol. 8, pp. 68514–68524, 2020.
- [10] Z. Xia, X. Wang, H. Wei, and Y. Xu, "Detection of vital signs based on variational mode decomposition using FMCW radar," in *Proc. Int. Conf. Microw. Millimeter Wave Technol.*, Nanjing, China, 2021, pp. 1–3.
- [11] Y. Wang, W. Wang, M. Zhou, A. Ren, and Z. Tian, "Remote monitoring of human vital signs based on 77-GHz mm-Wave FMCW radar," *Sensors*, vol. 20, no. 10, pp. 2999–3021, 2020.
- [12] Z. Ling, W. Zhou, Y. Ren, J. Wang, and L. Guo, "Non-contact heart rate monitoring based on millimeter wave radar," *IEEE Access*, vol. 10, pp. 74033–74044, 2022.
- [13] M. Xiang, W. Ren, W. Li, Z. Xue, and X. Jiang, "High-precision vital signs monitoring method using a FMCW millimeter-wave sensor," *Sensors*, vol. 22, no. 19, pp. 7543–7566, 2022.
- [14] M. López, C. Palacios, J. Romeu, and L. Jofre-Roca, "In-cabin MIMO radar system for human dysfunctional breathing detection," *IEEE Sensors J.*, vol. 22, no. 24, pp. 23906–23914, Dec. 2022.
- [15] M. Arsalan, A. Santra, and C. Will, "Improved contactless heartbeat estimation in FMCW radar via Kalman filter tracking," *IEEE Sens. Lett.*, vol. 4, no. 5, May 2020, Art. no. 7001304.
- [16] L. Liu, J. Zhang, Y. Qu, S. Zhang, and W. Xiao, "mmRH: Noncontact vital sign detection with an FMCW mm-wave radar," *IEEE Sensors J.*, vol. 23, no. 8, pp. 8856–8866, Apr. 2023.
- [17] Y. Wang et al., "A novel non-contact respiration and heartbeat detection method using frequency-modulated continuous wave radar," *IEEE Sensors J.*, vol. 24, no. 7, pp. 10434–10446, Apr. 2024.
- [18] S. Jeon et al., "W-band MIMO FMCW radar system with simultaneous transmission of orthogonal waveforms for high-resolution imaging," *IEEE Trans. Microw. Theory Techn.*, vol. 66, no. 11, pp. 5051–5064, Nov. 2018.
- [19] X. Liang, J. Deng, H. Zhang, and T. Gulliver, "Ultra-wideband impulse radar through-wall detection of vital signs," *Sci. Rep.*, vol. 8, no. 1, pp. 13367–13387, Sep. 2018.
- [20] L. Qu, C. Liu, T. Yang, and Y. Sun, "Vital sign detection of FMCW radar based on improved adaptive parameter variational mode decomposition," *IEEE Sensors J.*, vol. 23, no. 20, pp. 25048–25060, Oct. 2023.
- [21] M. Torres, M. Colominas, G. Schlotthauer, and P. Flandrin, "A complete ensemble empirical mode decomposition with adaptive noise," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Prague, Czech Republic, 2011, pp. 4144–4147.
- [22] T. Sakamoto, "Noncontact measurement of human vital signs during sleep using low-power millimeter-wave ultrawideband MIMO array radar," in *Proc. IEEE Int. Microw. Biomed. Conf.*, Nanjing, China, 2019, pp. 1–4.
- [23] W. Wang, C. Chen, W. Yao, K. Sun, W. Qiu, and Y. Liu, "Synchrophasor data compression under disturbance conditions via cross-entropy-based singular value decomposition," *IEEE Trans. Ind. Inform.*, vol. 17, no. 4, pp. 2716–2726, Apr. 2021.
- [24] P. Comon, "Independent component analysis, a new concept?," *Signal Process.*, vol. 36, no. 3, pp. 287–314, 1994.
- [25] J. Liu, H. Song, H. Sun, and H. Zhao, "High-precision identification of power quality disturbances under strong noise environment based on FastICA and random forest," *IEEE Trans. Ind. Informat.*, vol. 17, no. 1, pp. 377–387, Jan. 2021.
- [26] A. Hyvarinen, "Fast and robust fixed-point algorithms for independent component analysis," *IEEE Trans. Ind. Electron.*, vol. 10, no. 3, pp. 626–634, May 1999.
- [27] M. Prokhorov, V. Ponomarenko, V. Gridnev, M. Bodrov, and A. Bespyatov, "Synchronization between main rhythmic processes in the human cardiovascular system," *Phys. Rev. E*, vol. 68, no. 4, 2003, Art. no. 041913.
- [28] Texas Instrument, "AWR1642 Single-Chip 77 and 79GHz FMCW radar sensor," Texas Instrum., Dallas, TX, USA. (n.d.). [Online]. Available: <https://www.ti.com/lit/ds/symlink/awr1642.pdf>
- [29] Heal Force, "Prince-180B," 2020. [Online]. Available: <http://www.healforce.com/cn/index.php?ac=article&at=read&did=457>
- [30] HUAKE, "HKH-11C respiratory wave sensor," 2020. [Online]. Available: <http://www.hfuake.com/chanpinzhanshi/yiliaochuanganganqi/huxichuangangqi/2015/0101/hkh-11c.html>



Yong Wang (Member, IEEE) received the Ph.D. degree in information and communication engineering from the Harbin Institute of Technology, Harbin, China, in 2018.

He joined the Chongqing University of Posts and Telecommunications, Chongqing, China, where he is currently an Associate Professor. His research interests include millimeter wave sensing for vital signs and gesture recognition, machine learning, and quantum image processing.



Heng Liu received the M.S. degree in communications engineering from the Chongqing University of Posts and Telecommunications, Chongqing, China, in 2024.

His current research interests include mmWave radar vital signs detection technology.



Mu Zhou (Senior Member, IEEE) received the B.S., M.S., and Ph.D. degrees in information and communication engineering from the Harbin Institute of Technology (HIT), Harbin, China, in 2006, 2008, and 2012, respectively.

He is currently a Professor with the Chongqing University of Posts and Telecommunications, Chongqing, China. His research interests include wireless localization and sensing, signal processing and detection, and machine learning.



Wei Xiang (Senior Member, IEEE) received the Ph.D. degree in telecommunications engineering from the University of South Australia, Adelaide, SA, Australia, in 2004.

He is currently a Cisco Research Chair of AI and IoT and the Director of the Cisco-La Trobe Centre for AI and IoT, La Trobe University, Melbourne, VIC, Australia. His research interests include the Internet of Things, wireless communications, machine learning for IoT data analytics, and computer vision.



Yu Pang received the Ph.D. degree in electrical engineering from the Department of Electrical and Computer Engineering, McGill University, Montreal, QC, Canada, in 2010.

He is currently a Professor with Chongqing University of Posts and Telecommunications (CQUPT), Chongqing, China. His current research interests include integrated circuit design, vital signs detection, and artificial intelligence.



Jiacheng Wang received the Ph.D. degree in information and communication engineering from the School of Communications and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing, China, in 2022.

He is currently a Postdoctoral Research Fellow of College of Computing and Data Science with Nanyang Technological University, Singapore. His research interests include wireless sensing, semantic communications, generative

AI, and metaverse.



Dusit Niyato (Fellow, IEEE) received the Ph.D. degree in electrical and computer engineering from the University of Manitoba, Winnipeg, MB, Canada, in 2008.

He is currently a Professor with the College of Computing and Data Science, at Nanyang Technological University, Singapore. His research interests include sustainability, edge intelligence, decentralized machine learning, and incentive mechanism design.